



ing of the circuit boards maintains torque and facilitates rapid movement of magnetic fields for high-speed rotation. Each component is individually built, tested, and proven to exceed normal operational demands, minimising risks and ensuring reliable performance.

Lasers deployed to assess fracture resistance of 3D-printed cement

Engineers at Princeton University are deploying lasers to assess the fracture resistance of 3D-printed cement. Controlling laser power and speed for precise groove control allows



Princeton researchers work to improve the use of additive manufacturing for concrete and other cement-based materials. The research team, from left to right, Arjun Prihar, Reza Moini, Shashank Gupta, and Hadi Esmaeeli (Photo by Sameer Khan/Fotobuddy)

accurate testing. Additive technology creates robust materials through innovative design. 3D-printed concrete uses nozzles for gradual layer build-up. Moisture films during extrusion cause defects, weakening structure.

The researchers analysed interfaces to study their impact on fracture properties. The customised 3D printer is used for precise fabrication and curing, avoiding imprecise notching. This approach aids in determining material properties for scaling up concrete additive manufacturing in various applications. The researchers explored architecture materials, advancing their understanding of 3D-printed concrete fracture mechanics.

Novel method continuously monitors noise surrounding a quantum system

Researchers at the University of Chicago's Pritzker School of Molecular Engineering have developed a novel approach for continuously monitoring noise surrounding a quantum system.



In a new paper in *Science*, researchers with the lab of Asst. Prof. Hannes Bernien at the University of Chicago's Pritzker School of Molecular Engineering (above) describe a method to constantly monitor the noise around a quantum system and adjust the qubits, in real-time, to minimise error (Photo Credit: John Zich)

This method allows immediate qubit adjustments to minimise errors. The sensitivity of qubits to their surroundings leads to information alteration and higher error rates. Additionally, when a qubit is measured to evaluate noise,

its state collapses, leading to data loss. The researchers have achieved a hybrid atomic quantum processor utilising rubidium and caesium atoms. The team has created a system that reads real-time data from rubidium atoms and adjusts caesium atoms using microwave oscillations. The team exposed the quantum array to magnetic field noise to verify the minimisation of errors. Upon detecting the presence of caesium atoms, the system promptly eliminated the rubidium atoms.

Researchers believe only quantum error correction can help surpass classical supercomputers' capabilities

The researchers at IBM have simulated the behaviour of magnetic material on their Eagle quantum processor. Significantly, they overcame the challenge of quantum noise, which is the primary hurdle for this technology as it introduces calculation errors and impedes reliable results. The researchers have encoded qubits in superconducting circuits. The team measured qubit noise, accounting for predictable patterns due to positioning and fabrication imperfections. Extrapolating from noise-free conditions, they conducted calculations on all 127 qubits of Eagle, surpassing previous quantum computing experiments with 60 processing steps. The team aims to transition to quantum error correction, which will necessitate a significant increase in the number of qubits for each data qubit. The researchers doubted noise mitigation's potential and believed that only quantum error correction could achieve calculations surpassing classical supercomputers' capabilities.

New superconducting diode to expand quantum computers and enhance AI performance

Researchers at the University of Minnesota Twin Cities have created a superconducting diode. This significant advancement in electronic device components enables quantum computer expansion and enhances artificial intelligence performance. The device surpasses other superconducting diodes in energy efficiency, multitasking capability, and integration of energy flow control gates. The device uses three Josephson junctions, sandwiching superconductors with non-superconducting materials. The researchers established connections between the superconductors using semiconductor layers, allowing voltage-based control of their behaviour. The device showcases near-optimal energy efficiency and introduces gates and electric fields for fine-tuning. It utilises materials compatible with industrial processes, enhancing versatility and user-friendliness. This compatibility makes it suitable for industrial applications and enables scaling up quantum computer development for broader use.